Eliminating $a_0$, we can write the second equation as

$$0 = (1 - J_0)(1 - a_1) + hJ_0'(b_{-1} - b_1)$$

and the fourth equation as

$$0 = -J_2(1 - a_1) + hJ_2'(b_{-1} - b_1)$$

From these we see that

$$a_1 = 1$$
$$b_{-1} = b_1$$

and we are on the edge of stability. From the first equation

$$a_0 = 0$$

The formula is, therefore, of the form

$$y_{n+1} = y_{n-1} + b_1(y'_{n+1} + y'_{n-1}) + b_0 y'_n$$

The third equation is now [where we use the fact that $J_1'(0) = 1/2$]

$$2J_1 = 2hb_{-1}J_1' + \frac{hb_0}{2}$$

As $h \to 0$, this approaches

$$2b_{-1} + b_0 = 2$$

as it should since for a straight line (dc) it is correct and it includes Simpson's formula.

This result brings us back to Tick's method mentioned earlier in Sec. 34.6 Tick asked himself the question, "What integration formula should I use if I want the error curve in the frequency domain to be as small as possible over the lower half of the Nyquist interval?" He added the condition that the formula be exact if $y' =$ constant or (what is the same thing) if $y(x)$ is a straight line. Thus he required the symmetric formula to be of the form

$$y_{n+1} = y_{n-1} + h(b_1 y'_{n+1} + b_0 y'_n + b_1 y'_{n-1})$$

and to satisfy

$$2 = b_0 + 2b_1$$

He then proceeded to use a computer to compute the error curve for various choices of the parameter $b_0$. By a sequence of trials he came to the value

$$b_0 = 1.2832$$

and the formula we analyzed in Sec. 36.3.

By our analytical method we use both

$$2J_1 = 2hb_{-1}J_1' + \frac{hb_0}{2}$$

and the condition Tick used, namely

$$2 = b_0 + 2b_1$$

to get

$$b_{-1} = \frac{2J_1(h) - h}{[2J_1'(h) - 1]/h}$$

Tick used half the Nyquist interval, that is, 4 samples per cycle, or what is the same thing $h = \pi/2$. For this value of $h$

$$b_{-1} = 0.35785 \cdots$$

Compared to Tick's empirical result of $b_{-1} = 0.3584$ (difference $= 0.00055$) this is very good since we have not done the final leveling of the Chebyshev error curve as he did. On the other hand, we have a result for *all* step sizes, or what is the same thing, for all fractions of the whole Nyquist interval.

## 36.8 SUMMARY

In this chapter we have examined a number of integration formulas from the point of view of what they do to all the different frequencies. In practice it is often possible to connect such information with the physical problem. Thus, before an integration is begun, the proposer often has some idea of the frequency content of the solution that he thinks is important, and after the problem is done, he can relate the step size used to the highest frequency that is present to any large extent in the solution. Of course there are problems in which the frequency content has no physical meaning and this approach is not relevant.

From the new method of looking at the characteristics of a formula we have passed to one type of formula design, namely the requirement that the error curve in the frequency domain be exactly zero at dc and have a Chebyshev characteristic in some subpart of the total Nyquist interval. This is a very natural requirement in many applications.

# 37

# *DESIGN OF DIGITAL FILTERS

## 37.1 BACKGROUND

In processing analog signals, especially those in the form of electrical currents, it is usually necessary to change the amount of various frequencies present in the signal. In particular, it is often necessary to pass certain frequencies and to block others, to filter out certain frequencies. As a result of the importance of this problem there is a large, highly developed field of analog filter design.

When we digitize a signal (for machine processing or for more reliable transmission over noisy communication lines), we inherit the same filtering problems, but now we must filter out frequencies from a stream of digits. At first this problem of digital filtering was approached via the earlier field of analog filters, and if you already know a great deal about analog filters, this is a reasonable approach. But if you know almost nothing about this field, then the direct approach is more practical, and it is the one we will adopt as an introduction to digital filters. However, much can be learned from the older analog methods.

Digital filtering is not only easily done on a general-purpose computer, but special high-speed multipliers[1] and adders make it easy and economical in special applications.

## 37.2 A NONRECURSIVE CLASS OF DIGITAL SMOOTHING FILTERS

We begin the study of digital filters by recalling those in Sec. 35.5. There we studied smoothing and designed one simple filter that smoothed by 3s and then by 4s and which did a very good job at blocking out the upper half of the frequency spectrum (a low-pass filter). We also examined the filtering action of two least-squares smoothing formulas, and one Chebyshev formula.

These examples suggest that we study the general family of 5-term linear formulas (filters) of the form

$$g_k = af_{k-2} + bf_{k-1} + cf_k + df_{k+1} + ef_{k+2}$$

However, in this section we shall limit ourselves to *symmetric* formulas ($e = a$, $d = b$) both because they are the ones that are widely used in practice and because for smoothing they have no phase distortion (no imaginary terms) and this makes them easier to understand. Later (Sec. 37.4) we will examine differentiation formulas which have a pure imaginary phase component. We therefore first examine the restricted class of symmetric smoothing formulas

$$g_k = af_{k-2} + bf_{k-1} + cf_k + bf_{k+1} + af_{k+2}$$

As usual, we start by asking what happens to a single frequency by assuming that

$$f(t) = e^{2\pi i\sigma t} = e^{i\omega t}$$

or

$$f_k = e^{i\omega k}$$

Thus we get

$$g_k = (2a \cos 2\omega + 2b \cos \omega + c)f_k$$

The transfer function $H(\omega)$ which multiplies the amplitude of the input signal of frequency to get the output amplitude is

$$H(\omega) = 2a \cos 2\omega + 2b \cos \omega + c$$

Without the even symmetry we would have had some sine terms with imaginary coefficients. In some of the previous work we used $H(\omega)$ as the ratio of the computed to true, and for smoothing $g_k/f_k$ is this ratio, and so our notation is consistent.

[1] 40 ns, 17-bit multipliers in 1971.

How shall we choose the filter coefficients $a$, $b$, and $c$? That depends, of course, on what we want to do. Suppose that again we want to block out the high frequencies and pass the low frequencies (a low-pass filter). To convert this to passing the high and blocking the low frequencies (a high-pass filter), we have only to compute $f_k - g_k$ as a new filter.

We will start by arbitrarily imposing two conditions on the frequency curve, namely that at the low end

$$H(0) = 1 \qquad \text{full dc (lowest frequency)}$$

and at the upper end

$$H(\pi) = 0 \qquad \text{no highest frequency}$$

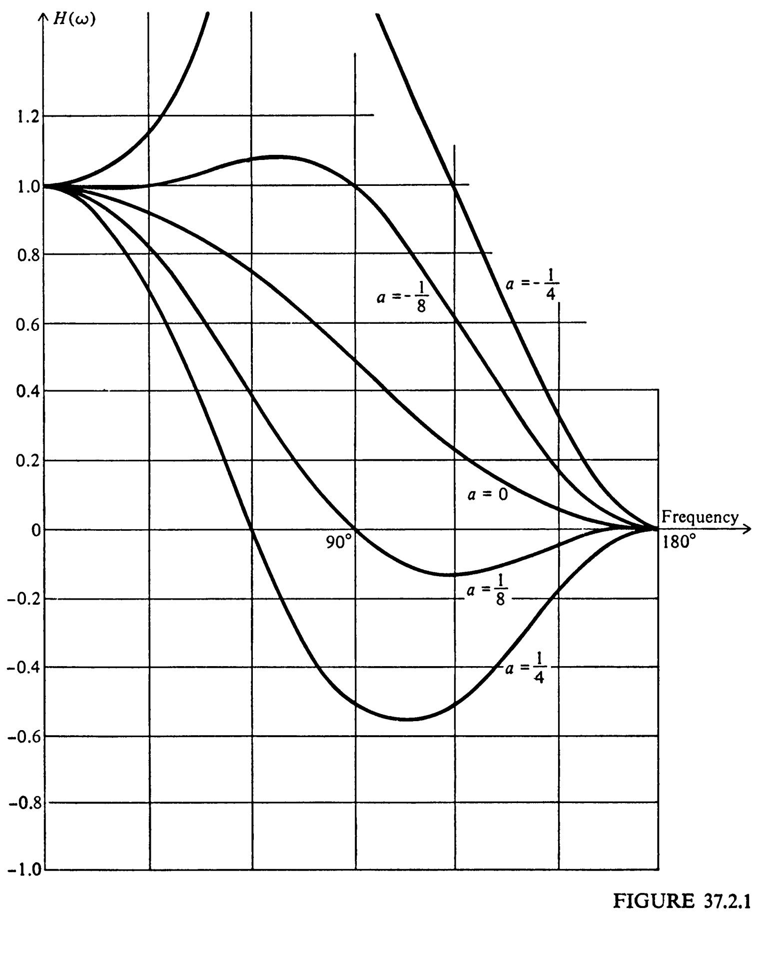


FIGURE 37.2.1

These two conditions mean

$$H(0) = 2a + 2b + c = 1$$
$$H(\pi) = 2a - 2b + c = 0$$

From these we get

$$b = 1/4$$
$$c = 1/2 - 2a$$

and we have a one-parameter family of filters (see Fig. 37.2.1).

$$H(\omega) = 2a \cos 2\omega + 1/2 \cos \omega + (1/2 - 2a) = 4a(1 + \cos \omega)\left[\cos \omega - \left(1 - \frac{1}{8a}\right)\right]$$

Note that $H(\omega)$ is a periodic even function of $\omega$ and that it is also symmetric about $\omega = \pi$.

From this figure we can pick out a filter that approximately meets our needs. Alternately, we could impose one more condition on the transfer function $H(\omega)$ and directly determine the filter.

As a first example of directly determining the filter, suppose we require

$$H_1\left(\frac{\pi}{2}\right) = 1$$

Thus we get

$$H_1\left(\frac{\pi}{2}\right) = -2a + 1/2 - 2a = 1$$
$$a = -1/8$$
$$c = \quad 3/4$$

Therefore

$$g_k = 1/8(-f_{k-2} + 2f_{k-1} + 6f_k + 2f_{k+1} - f_{k+2})$$

and

$$H_1(\omega) = -\frac{\cos 2\omega}{4} + \frac{\cos \omega}{2} + \frac{3}{4}$$

As a second example of the design of a filter, suppose we balance the two halves of the filter by setting

$$H_2\left(\frac{\pi}{2}\right) = 1/2$$

Then we get

$$H_2\left(\frac{\pi}{2}\right) = -2a + 1/2 - 2a = 1/2$$
$$a = 0$$
$$c = 1/2$$
$$g_k = 1/4(f_{k-1} + 2f_k + f_{k+1})$$
$$H_2(\omega) = \frac{\cos \omega}{2} + \frac{1}{2}$$

As a third example, suppose we try to do as well as we can in the neighborhood of dc (zero frequency). We already have both

$$H_3(0) = 1$$

and

$$\frac{dH_3(0)}{d\omega} = 0$$

hence we impose the further condition that

$$\frac{d^2H_3(0)}{d\omega^2} = 0 = -8a - 1/2$$
$$a = -1/16$$
$$c = 5/8$$
$$g_k = \frac{1}{16}(-f_{k-2} + 4f_{k-1} + 10f_k + 4f_{k+1} - f_{k+2})$$
$$H_3(\omega) = -\frac{\cos 2\omega}{8} + \frac{\cos \omega}{2} + \frac{5}{8}$$

As a final example, suppose we ask that in the lower half of the Nyquist interval $0 \le \omega \le \pi/2$, $H(\omega)$ have the shape of a Chebyshev approximation to unity, while still keeping the two conditions $H_4(0) = 1$ and $H_4(\pi) = 0$. The position of the extreme value of $H_4(\omega)$ is defined by

$$\frac{dH_4(\omega)}{d\omega} = -4a \sin 2\omega - \frac{1}{2}\sin \omega = 0$$

This defines the value $\omega_0$ at which the extreme value occurs. To find $\omega_0$, we use the double-angle formula for $\sin 2\omega$ to get

$$-8a \sin \omega \cos \omega - \frac{1}{2}\sin \omega = 0$$
$$\sin \omega = 0 \qquad \omega = 0, \pi$$
$$\cos \omega = -\frac{1}{16a}$$

Hence the extreme value is

$$H_4(\omega_0) = 2a \cos 2\omega_0 + \frac{1}{2}\cos \omega_0 + \frac{1}{2} - 2a$$
$$= \frac{1}{2} - 4a - \frac{1}{64a}$$

The value at $\omega = \pi/2$ is

$$H_4\left(\frac{\pi}{2}\right) = 1/2 - 4a$$

The excess above 1 at $\omega_0$ is to be equated to the amount below at $\pi/2$, since the Chebyshev condition means that

$$H_4(\omega_0) - 1 = 1 - H_4\left(\frac{\pi}{2}\right)$$

This leads to the equation

$$2^9 a^2 + 2^6 a + 1 = 0$$

whose solution is

$$a = \frac{-2 \pm \sqrt{2}}{32}$$

Thus

$$\cos \omega_0 = \frac{-1}{16a} = \frac{-2}{-2 \pm \sqrt{2}} = 2 \pm \sqrt{2}$$

Evidently we must use the minus sign (if we want a real $\omega_0$), and

$$a = \frac{-2 - \sqrt{2}}{32} = \frac{-3.414}{32} = -0.1067 \cdots$$
$$c = 0.7134$$

Looking at the figure we see that this is about the expected value.

We have thus shown a number of simple ways we can design a low-pass filter.

## 37.3 AN ESSAY ON SMOOTHING

We have just examined some low-pass digital filters. We again note that by merely taking $f_k - g_k$ we can get a high-pass filter that accepts what the previous one rejected and rejects what the previous one accepted. Thus we have a high-pass filter corresponding to each low-pass filter in the previous section.

We earlier identified smoothing with the removal of high frequencies on the grounds that random roundoffs are like small jumps in the function which give rise to high frequencies as well as low frequencies, while the signal we are usually dealing with has mainly low frequencies. We further showed (Sec. 35.2) that the difference-table method of measuring the noise amplified the upper two-thirds of the Nyquist interval and damped out the lower one-third.

It is not easy to state in a vacuum exactly what filter to use to smooth a given function. It depends on the spectra of *both* the signal and noise. Evidently we can build a filter to reject the frequencies that we think are noise and not signal, but we cannot, by any linear method of smoothing, reject the frequencies of the noise in the range of the signal *unless* we are prepared to lose some of the signal as well. The practical art of filtering to remove noise often includes this difficult compromise of removing a small amount of the signal to get rid of a lot of the noise as well. And to do this we need to have an accurate estimate of the spectrum of the signal as well as the spectrum of the noise.

In practical measurements it is often possible to form some judgments as to the respective spectra, but in computing all too little is known of the spectrum of the roundoff noise—more studies are required in this area. It is generally believed to be flat (white noise).

Viewed as smoothing filters, some of the formulas we have given violate a very natural condition, namely that if one function is uniformly larger than another, then the smoothed values should also be larger. This condition is formalized in the following theorem.

**Theorem 37.3.** If $u_k \geq v_k$ for all $k$, then a necessary and sufficient condition for the smoothed value

$$\bar{u}_k = \sum_{i=k-N}^{k+N} a_i u_i$$

to be greater than or equal to the corresponding smoothed value $\bar{v}_k$ is that all the

$$a_i \geq 0$$

PROOF If the $a_i$ are all positive, then the smoothed values are clearly in the right order (we have only to look at the smoothed difference of $u_k - v_k$). On the other side of the argument, if one of the $a_i < 0$, then consider $u_k = 0$ for all $k$ and $v_k = 0$ except for $v_i < 0$.
Then

$$\bar{v}_i > \bar{u}_i = 0$$

Hence the theorem is proved. ////

For the family in Sec. 37.2

$$a = a$$
$$b = 1/4$$
$$c = 1/2 - 2a$$

This means that $0 \leq a \leq 1/4$ ($1/2 \geq c \geq 0$), and many of the interesting filters are eliminated by this condition.

This theorem should not be taken as gospel. Consider Fig. 37.3.1 where $u_k \equiv 0$ (circles) and where $v_k$ is as shown (crosses). It is *not* unreasonable to have the smoothed middle value of $v_k > 0$ as shown by the curve.

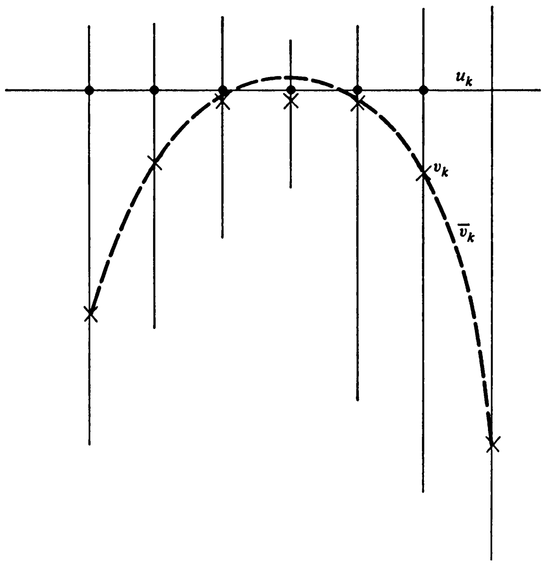


FIGURE 37.3.1

## 37.4 DIFFERENTIATION FILTERS

Differentiation, even more than smoothing, is a delicate process, since it tends to amplify the high frequencies (the noise in the signal). For differentiation we use a filter with odd symmetry ($c = -a$, $d = -b$, $c = 0$)

$$g_k = af_{k+2} + bf_{k+1} - bf_{k-1} - af_{k-2}$$

The usual substitution

$$f_k = e^{i\omega k}$$

gives

$$g_k = i(2a \sin 2\omega + 2b \sin \omega) f_k$$

where of course the derivative (with respect to $k$) is

$$f_k' = i\omega f_k \simeq g_k$$

Thus the parentheses should be an approximation to $\omega$ since the transfer function is the ratio of the computed to the true

$$H(\omega) = \frac{2a \sin 2\omega + 2b \sin \omega}{\omega}$$

We cannot approximate it in the whole interval, for it is clear that the term in the parentheses is zero at $\omega = \pi$. We shall impose the condition

$$H(0) = 1$$

This means that as $\omega \to 0$,

$$4a \cos 2\omega + 2b \cos \omega \to 1$$

$$4a + 2b = 1$$

or

$$b = \frac{1}{2}(1 - 4a)$$

and

$$g_k = a(f_{k+2} - f_{k-2}) + \frac{1-4a}{2}(f_{k+1} - f_{k-1})$$

$$H(\omega) = \frac{2a \sin 2\omega + (1 - 4a) \sin \omega}{\omega}$$

$$= \frac{\sin \omega}{\omega}[1 - 4a(1 - \cos \omega)]$$

Thus we have a one-parameter family of differentiation formulas to explore.

As our first example, we ask that

$$H\left(\frac{\pi}{2}\right) = 1 = \frac{1 - 4a}{\pi/2}$$

hence

$$a = \frac{1}{4}\left(1 - \frac{\pi}{2}\right)$$

$$b = \frac{\pi}{4}$$

As a second example, suppose we try to do as well as we can in the neighborhood of $\omega = 0$, that is, $H'(0) = 0$. This becomes

$$H(\omega) = \frac{1}{\omega}\left\{2a\left[2\omega - \frac{(2\omega)^3}{3!} + \frac{(2\omega)^5}{5!} - \cdots\right] + (1 - 4a)\left[\omega - \frac{\omega^3}{3!} + \frac{\omega^5}{5!} - \cdots\right]\right\}$$

$$= 1 + \omega^2\left(\frac{-8a}{3} - \frac{1}{6} + \frac{2a}{3}\right) + \omega^4\left(\frac{8a}{15} + \frac{1}{120} - \frac{a}{30}\right) + \cdots$$

We equate the coefficient of $\omega^2$ to zero to get $a = -1/12$, which leads to

$$H_2(\omega) = 1 - \frac{\omega^4}{30} + \cdots$$

$$g_k = -\frac{1}{12}[f_{k+2} - f_{k-2}] + \frac{2}{3}[f_{k+1} - f_{k-1}]$$

We can similarly design other filters to meet other conditions.

## PROBLEMS

*37.4.1* Design a filter for the second derivative.

*37.4.2* Design a Chebyshev in $0 \leq \omega \leq \pi/2$ filter for a first derivative. [Use $H(\omega)$.]

## 37.5 RECURSIVE FILTERS

Nonrecursive filters use only the input function values and do not make use of already computed output numbers, while recursive filters use both. There are occasions when, as in the two integration formulas for the trapezoid and midpoint rules (Secs. 35.4 and 36.2), both methods give the same result. For a given span (number of back terms used in the formula) the nonrecursive formula has a limited memory of the past, while a recursive formula, in a certain sense, remembers all the past.

For a given span the recursive formula will have almost twice as many parameters as the nonrecursive formula, and hence it can be made to fit a desired frequency response far better. However, it should be remembered that the recursive formula will have to be watched for stability problems.

The predictor-corrector methods for integrating ordinary differential equations that we developed using polynomial approximation, and later in Chap. 36 analyzed in the frequency domain, are good examples of recursive formulas. They also illustrate the situation where we tend to use recursive filters, namely when we have information only up to the present and we need to take some action or make some prediction of what is going to happen next (extrapolation). Nonrecursive filters tend to be used when all the data are available at the time we are processing them.

The three classical operations of filters are integration, smoothing (interpolation, extrapolation), and differentiation, but in principle filters apply to any linear operation on the data.

The recursive filters have another property worth mentioning. In the integration of differential equations, when we want great accuracy, we use at least 10 samples in the highest frequency present, and even for rough work we use at least 5 samples. But in many data processing problems, where the noise is high, we cannot afford that luxury (because among other reasons the spectrum of the noise for high frequencies is too large). Thus we find that we are occasionally working around the Nyquist rate of 2 samples in the highest frequency present (to any extent).

The topic of filters is of great importance in many digital problems of signal processing, such as the Moon and Mars pictures, and whole books[1] are currently devoted to the topic of digital filters, so we must (reluctantly) refer the reader to them.

[1] The classic reference is F. F. Kuo and J. F. Kaiser [32], especially the chapter by Kaiser. A more modern reference is B. Gold and C. M. Rader [17]. A further important reference with respect to practical computing is V. V. Solodovnikov [54]. Also *IEEE Transactions on Audio and Electroacoustics*, vol. AU-16, no. 3, September, 1968.

# 38

# *QUANTIZATION OF SIGNALS

## 38.1 INTRODUCTION

We have already examined the effects of sampling an analog signal in the time (independent) variable. The process of sampling also replaces the continuous analog signal by a digital output that can be only one of comparatively few numbers. Thus the signal is not only sampled in the independent variable, it is also *quantized* in the dependent variable. Since digital computing involves a great deal of processing of quantized physical data, it is necessary to look carefully at this topic. (See Fig. 38.1.1.)

The quantization is similar to roundoff *except* that we normally think of roundoff as occurring in relatively high precision numbers, while the quantization effect in measured signals may often be severe. The process of quantization occurs in what is usually called analog-to-digital conversion, or $A$-to-$D$ conversion. This process may have precision anywhere from 1 to 17 bits (possibly more in a few cases), so that quantization may be very important or may be a relatively small effect. In many situations a 1-bit digitizer can give a surprising amount of information (Sec. 38.6).

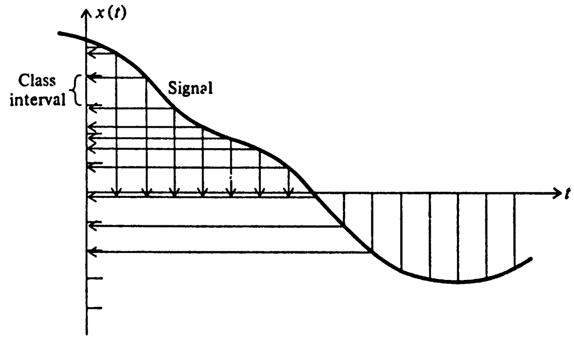


FIGURE 38.1.1

The topic of quantization is a large one and we can give only a brief introduction, referring the interested reader to the literature [17, 20, 58] for more details; in a sense we are only trying to create an awareness of the consequences of these widely ocurring effects.

## 38.2 THE GRAY CODE

The analog to digital converter usually does not go from the analog signal directly to the binary representation of the quantized number, rather there is an intermediary step of representation in a form known as *the Gray code*.[1] The reason for this is easily understood if we try to imagine how the converter works. It is convenient to think of a rotating wheel with sectors, and we are trying to read optically the encoding on the sector under the reading heads. If there is a sector directly under the heads at the moment of reading, there will be little trouble, but suppose the reading heads are partly over one sector and partly over another sector, in particular between sectors 7 and 8. (See Fig. 38.2.1. In binary form, this is:

0 111
1 000

Clearly, almost anything between 0 000 and 1 111 (0 and 15) might be read out.

[1] Technically *a* Gray code, but this one is so universally used that it is *the* Gray code.

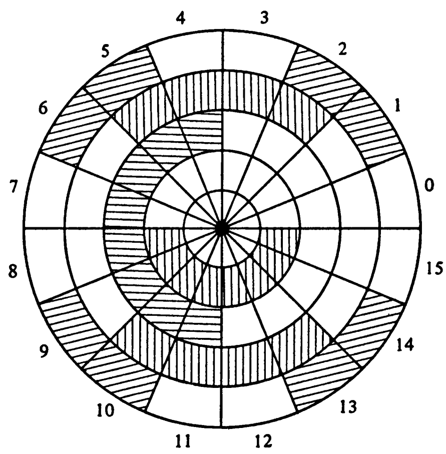


FIGURE 38.2.1

The Gray code has the important property that in going from one number to the next only one digit changes, thus eliminating this awkward problem of reading parts of two sectors at the same time and then getting almost any number. Because only one digit changed, one of the two possible numbers must come out.

We shall first show how to construct a Gray code. The code using 1 bit is clearly

| | |
|---|---|
| 0 | 0 |
| 1 | 1 |

and note that it is really circular and the bottom is followed by the top of the list. To form the 2-bit code we put a 0 in front of each to get

00
01

We also copy the original code below this in the *reverse* order and put 1s in front of this, getting, finally, the 2-bit Gray code

| | |
|---|---|
| 0 | 00 |
| 1 | 01 |
| 2 | 11 |
| 3 | 10 |

At each stage we get the next larger code by first copying the earlier code with an added 0 in front and then copying the original code in reverse order with 1s in front. For example, in going from a 3-bit to a 4-bit code we get

| | |
|---|---|
| 0 | 0 000 |
| 1 | 0 001 |
| 2 | 0 011 |
| 3 | 0 010 |
| 4 | 0 110 |
| 5 | 0 111 |
| 6 | 0 101 |
| 7 | 0 100 |
| | |
| 8 | 1 100 |
| 9 | 1 101 |
| 10 | 1 111 |
| ⋮ | ⋮ |
| 15 | 1 000 |

The characteristic property of the Gray code follows immediately from the method of construction.

It sometimes happens that the conversion from the Gray code to binary number code is not done in the converter, and therefore it must be done in the computer. We now describe how this is done. A study of the method of construction shows that:

*1* The left-hand digit is the correct binary digit.

*2* If the leftmost digit is 0, then the next digit is correct; but if it is a 1, then the next digit should be complemented.

*3* In general, if at some stage in the decoding there has been an odd number of 1s in the Gray-code representation, then the next digit must be complemented; but if there has been an even number of 1s, then the next digit is correct.

The rare case of having to convert the other way (it can arise in a simulation of a system which has converters) is left as an exercise.

## PROBLEMS

*38.2.1* Give the equations for determining the digits of the Gray-code representation of a binary number.

*38.2.2* Sketch a 5-bit Gray-code wheel.

## 38.3 THE STATISTICAL DISTRIBUTION OF VALUES

Since quantization is very nonlinear, it is not possible to give a simple analysis of what happens, and we will examine only the *statistics* of the process.

We begin with an examination of the distribution of the values of the function. We want to know how many times a given value occurs. To get this we need to project the curve horizontally onto the vertical axis and note how many values end up at a given point along this axis, for both the discrcte and the continuous cases. See Fig. 38.3.1.

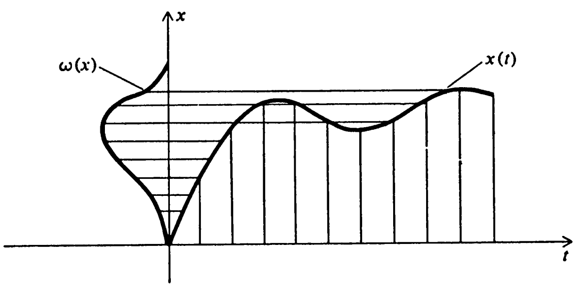


FIGURE 38.3.1

As an example of this process, consider the distribution of the values of a single sinusoid $x = \sin t$. See Fig. 38.3.2. Clearly we need to examine only the interval $-\pi/2$ to $\pi/2$. The area under the density curve $x(w)$ between $x$ and $x + dx$ is the fraction of the time the signal $x(t) = \sin t$ is between $x$ and $x + dx$, that is,

$$dt = w(x)\, dx$$

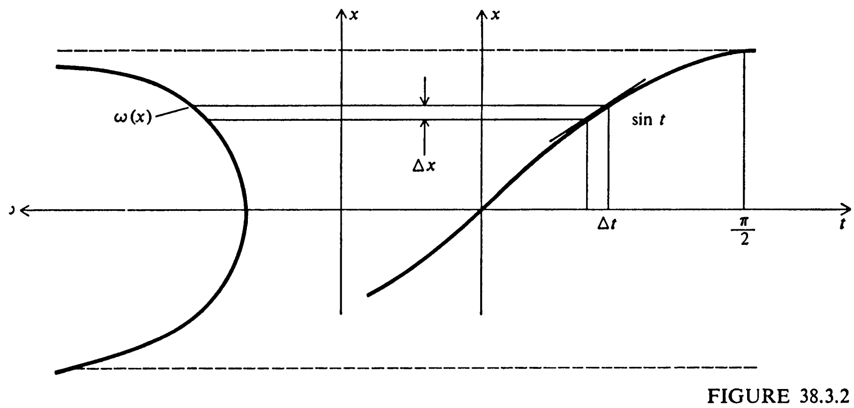


FIGURE 38.3.2

Since $x = \sin t$, then

$$\frac{dx}{dt} = \cos t$$

or

$$dt = \frac{dx}{\cos t}$$

Comparing the two expressions for $dt$, we find

$$w(x) = \frac{1}{\cos t} = \frac{1}{\sqrt{1 - \sin^2 t}} = \frac{1}{\sqrt{1 - x^2}}$$

To convert this distribution to a probability distribution, we need to normalize it so that

$$\int_{-1}^{1} w(x)\, dx = 1$$

But we have

$$\int_{-1}^{1} \frac{dx}{\sqrt{1 - x^2}} = \arcsin x \Big|_{-1}^{1} = \pi$$

Hence we must use

$$w(x) = \begin{cases} \dfrac{1}{\pi} \dfrac{1}{\sqrt{1 - x^2}} & -1 \le x \le 1 \\ 0 & \text{elsewhere} \end{cases}$$

in place of the old $w(x)$.

In the general case of $x = x(t)$ we use

$$\frac{dx}{d[x(t)]/dt}$$

We can either normalize the probability range before we start (note in the above example the range is $\pi$, the normalizing factor) or else normalize when we are done.

It is this function $w(x)$ that leads to the statistics of the analog-to-digital conversion process. Notice in the case of $x = \sin t$ that the "favorite values" are at the extremes of the range and not in the middle.

## PROBLEMS

Find $w(x)$ for:

*38.3.1* $x = e^{-t}$ $0 \le t \le N$, and 0 elsewhere

*38.3.2* A circle

*38.3.3* $x(t) = \tan t$ $(0 \le t \le \pi/4)$, 0 elsewhere

## 38.4 NOISE DUE TO QUANTIZATION

When we studied the roundoff noise, the precision of the numbers was such that it was reasonable to assume that the roundoff errors were uniformly distributed in the interval, and any skewness of the distribution of the numbers (see Secs. 2.8 and 2.9) did not reach that far down. But when the quantization is imprecise, then the skewness of the distribution of the values can affect the statistics of the quantization (roundoff). We therefore examine this effect, which turns out to be the well-known Sheppard's corrections (which we now derive).

We *assume* that the distribution function of the values of $w(x)$ has the property that the derivatives

$$w'(x),\ w''(x),\ w'''(x), \ldots$$

all approach zero as the value of $x$ gets large in either direction, and also that

$$x^k w'(x),\ x^k w''(x), \ldots$$

approach zero for finite $k$. Usually this is true for distribution functions, and so it is not an unreasonable assumption.

In our quantization problem the range can be reduced to $a \le x \le b$ and divided into $n$ quantization (class) intervals of equal size that correspond to the equal-size steps of the usual analog-to-digital converter. The intervals will be

$$(x_i - q/2,\ x_i + q/2)$$

where $b - a = qn$.

Under these assumptions when we apply the Euler-Maclaurin, Gregory, or Hamming-Pinkham equal-weight integration formulas (Sec. 18.9), the correction terms will all vanish, and we will have

$$\int_{-\infty}^{\infty} x^r w^{(s)}(x)\, dx = q \sum_{i=1}^{n} x_j{}^r w^{(s)}(x_j)$$

The probability $p_j$ that we are in the $j$th quantization interval is

$$p_j = \int_{x_j - q/2}^{x_j + q/2} w(x)\, dx$$

and the $r$th moment is calculated from the quantized (grouped) data by

$$\mu_r'' = \sum_{j=1}^{n} x_j{}^r p_j$$

where the double primes mean calculated values, not derivatives. On the other hand the true value is, using a single prime for it,

$$\mu_r' = \int_a^b x^r w(x)\, dx$$

We can now expand the expression for the probability $p_j$

$$
\begin{aligned}
p_j &= \int_{x_j - q/2}^{x_j + q/2} w(x)\,dx = \int_{-q/2}^{q/2} w(x_j + x)\,dx \\
&= \int_{-q/2}^{q/2} \left[ w(x_j) + xw'(x_j) + \frac{x^2}{2} w''(x_j) + \frac{x^3}{3!} w'''(x_j) + \cdots \right] dx \\
&= qw(x_j) + \frac{q^3}{24} w''(x_j) + \frac{q^5}{1{,}920} w^{iv}(x_j) + \cdots
\end{aligned}
$$

The calculated $r$th moment is therefore, using our assumptions that led to the equivalence of the sum and integral,

$$
\begin{aligned}
\mu_r'' &= \sum_{j=1}^{n} x_j^r p_j \\
&= \int_a^b x^r w(x)\,dx + \frac{q^2}{24} \int_a^b x^r w''(x)\,dx + \frac{q^4}{1{,}920} \int_a^b x^r w^{iv}(x)\,dx + \cdots
\end{aligned}
$$

When integrating by parts, the integrated terms vanish, and we get

$$
\mu_r'' = \mu_r' + \frac{q^2}{24} r(r-1) + \frac{q^4}{1{,}920} r(r-1)(r-2)(r-3) + \cdots
$$

If the moments about the mean are $\mu$ (no primes), we get

$$
\begin{aligned}
\mu_0'' &= \mu_0 = 1 \\
\mu_1'' &= \mu_1 = 0 \\
\mu_2'' &= \mu_2 + \frac{1}{12} q^2 \mu_0 = \mu_2 + \frac{q^2}{12} \\
\mu_3'' &= \mu_3 + \frac{1}{4} q^2 \mu_1 = \mu_3 \\
\mu_4'' &= \mu_4 + \frac{q^2}{2} \mu_2 + \frac{q^4}{80}
\end{aligned}
$$

Solving the other way, we get the true (unprimed) moments in terms of the calculated (grouped, double-primed) moments

$$
\begin{aligned}
\mu_1 &= \mu_1'' = 0 \\
\mu_2 &= \mu_2'' - \frac{q^2}{12} \\
\mu_3 &= \mu_3'' \\
\mu_4 &= \mu_4'' - \frac{q^2}{2} \mu_2'' + \frac{7}{240} q^4
\end{aligned}
$$

......................

Evidently the correction in the second moment calculated from the quantized (grouped) data is particularly simple, and it is the main effect due to quantization that is used in practice—rarely do we go to the fourth moments.

## 38.5 THE QUANTIZATION THEOREM

Although we have expressed the quantized moments $\mu_r''$ in terms of the true moments $\mu_r$, and vice versa, there is still the question of when we can be sure that from the quantized data we can recover the true moments—the formalism may fail us if the series, for example, does not converge. This question is much like that answered by the sampling theorem which gives the conditions under which we can reconstruct the original function from the samples. There is a very similar theorem for quantization, based in the same way on the Fourier transform, but the physical meaning of the theorem is difficult to apply in practice, so we shall ignore it and proceed with an intuitive approach.

At first one might reason as follows. Let's suppose that the distribution of values $w(x)$ is reasonably normal. This assumption seems to fly in the face of the special case of $x = \sin t$ that we used as an example and where we found that the distribution of the values tended to pile up at the extremes. But for the moment we proceed anyway, holding on to our doubts. We could explain it, for example, by assuming that the extremes do *not* occur at the same level each time, and the result gives the normal distribution. One would be tempted in the case of, say, 8 levels of quantization to pick the step size to be $\sigma$. Thus most of the cases will fall in the range of the middle two intervals, and almost every value will fall in the $3\sigma$ range, and we still have a fourth to cover the rest, though there will be a little "clipping" when the value exceeds $4\sigma$.

It is clear what is wrong with the plan. We are using most of the intervals of quantization for almost none of the values, and this is stupid. But then probably so was the assumption of a normal distribution.

Thus it is necessary to look more closely at the distribution of the values $w(x)$. It has a "dynamic" range from the least to the maximum that must be covered, or else we must accept some clipping of the extreme cases. Once we have decided this problem, if the rest of the values do not fall rather uniformly in the range, then we must consider what is known as "companding"—compressing and later expanding. We look for a transformation of the dependent variable that will transform the distribution of values to make it fairly uniform. Then, using this distribution, we quantize into equal steps and get the numbers into the computer. Inside we have to first undo the compression (by using the inverse transformation to expand it). In this way we get the most information through the narrow bottleneck of the analog-to-digital converter with its limited range of numbers. Once inside the computer, we almost always have a far greater range available for the expanded range.

"Companding" in one form or another is widely used; for example, in both phonograph records and tapes the playing unit has "compensation circuits" (expanding) to correct for the deliberate recording distortion to keep a reasonable dynamic range.

## 38.6 THE POOR MAN'S FOURIER SERIES

When most people who work in Fourier series hear that a 1-bit quantizer can give a lot of information, they immediately think along the following lines. Suppose I were to quantize the trigonometric functions that occur in the computation of the coefficients of a Fourier series, how well would I do using only additions and subtractions of the original data? (Carried further, this idea leads to Walsh functions, etc.)

It is easy to analyze. We get the calculated coefficient values[1]

$$A_m = \frac{1}{\pi}\int_0^{2\pi} f(x)\,\mathrm{sign}(\cos mx)\,dx \qquad B_m = \frac{1}{\pi}\int_0^{2\pi} f(x)\,\mathrm{sign}\,(\sin mx)\,dx$$

In order to analyze exactly what these numbers are, we begin by analyzing the square wave

$$\mathrm{sign}(\sin x) = \begin{cases} 1 & 0 \le x < \pi \\ -1 & \pi \le x < 2\pi \end{cases}$$

This function has the Fourier-series coefficients (following the $\sin x$ only)

$$b_{1,k} = \frac{1}{\pi}\int_0^{2\pi} \mathrm{sign}(\sin x)\sin kx\,dx$$

which is zero for even frequencies and for odd frequencies is

$$b_{1,2k+1} = \frac{2}{\pi}\int_0^{\pi} \sin(2k+1)\,x\,dx = \frac{2}{\pi}\,\frac{\cos(2k+1)\,x}{-(2k+1)}\bigg|_0^{\pi} = \frac{4}{\pi(2k+1)}$$

Thus we have

$$\mathrm{sign}(\sin x) = \frac{4}{\pi}\sum_{k=0}^{\infty} \frac{\sin(2k+1)\,x}{2k+1}$$

Replacing $x$ by $mx$, we get

$$\mathrm{sign}(\sin mx) = \frac{4}{\pi}\sum_{k=0}^{\infty} \frac{\sin(2k+1)\,mx}{2k+1}$$

[1] The sign function is often written *sgn* (signum).

Thus

$$\begin{aligned}
B_m &= \frac{1}{\pi}\int_0^{2\pi} f(x)\frac{4}{\pi}\sum_{k=0}^{\infty}\frac{\sin(2k+1)\,mx}{2k+1}\,dx \\
&= \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{1}{2k+1}\frac{1}{\pi}\int_0^{2\pi} f(x)\sin(2k+1)\,mx\,dx \\
&= \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{1}{2k+1}\,b_{m(2k+1)}
\end{aligned}$$

where the $b_j$ are the true Fourier-series coefficients of $\sin jx$. These equations are triangular in form

$$\begin{aligned}
B_1 &= \frac{4}{\pi}\left\{b_1 + \frac{b_3}{3} + \frac{b_5}{5} + \cdots\right\} \\
B_2 &= \frac{4}{\pi}\left\{b_2 + \frac{b_6}{3} + \frac{b_{10}}{5} +\right\} \\
B_3 &= \frac{4}{\pi}\left\{b_3 + \frac{b_9}{3} + \cdot\ \cdot\right\}
\end{aligned}$$

etc.

If the original series converges rapidly, then we can solve these equations approximately for the $b_j$ in terms of the $B_m$, and the solution is quite stable. Similarly for the cosine-term coefficients $a_j$ in terms of the $A_m$.

In the case of a finite number of equally spaced points (and a finite number of terms in the Fourier expansion), we need clearly to use the midpoints $t_p$ and avoid the endpoints of the intervals $x_p$. Confusion begins if you try to go directly from the continuous to the quantized discrete; the two stages done separately limits the confusion.

## 38.7 SOME GENERAL REMARKS ON QUANTIZATION EFFECTS

Quantization effects are often not as serious as the beginner is apt to think. It depends a great deal on what the quantized signal is going to be used for. The reader has probably seen the effects of quantization of photographs into two, four, eight, and sixteen levels. Using 1 bit is simply going to the extreme of contrast in the developing stage of the photograph. Sec. 38.6 shows reasonably clearly how much two or four levels would affect a Fourier expansion. Thus simple two-level quantization can at times be used successfully.

On the other hand, for some purposes quantization can be very serious. Much depends on the consumer of the information. If the consumer is a human, who to some extent has been engineered by evolution to be able to comprehend signals in the presence of severe noise, quantization can leave the message still intelligible. But it is also true that if you want high fidelity, the human is again well engineered to detect very small differences.

If the consumer is not a human, but further formulas, then we need to consider the two extreme conditions of the previous paragraph. The mere detection of the presence of a signal may be easy, but the detailed information carried by the signal may be very hard to recapture if the quantization has been at all serious. Thus the *use* of the quantized signal is central to the question of how serious the quantization is.

# PART IV

# Exponential Approximation

# 39
# SUMS OF EXPONENTIALS

## 39.1 INTRODUCTION

When considering a basis for the representation of a function there are three classes of functions that occur naturally: polynomials, sines and cosines, and exponentials. All these have the property that if a function can be represented in terms of them, then it can also be so represented if the origin is shifted. This is not true, for example, of the class $1, 1/x, 1/x^2, \ldots, 1/x^n$. It is only true for the above three classes and combinations of them. The numerical integration of differential equations shows why this property is relevant.

We studied the first class, polynomials, in Part II (Chaps. 14 through 30) and the second class, sinusoids, in Part III (Chaps. 31 through 38), and now we come to the exponentials in Part IV (three short chapters). The reasons for this disparity of treatment are several. First, polynomials are indeed more useful in many situations because they are also invariant under a stretch of $x \to kx'$ (because a polynomial of degree $n$ in $x$ is still a polynomial of degree $n$ in $kx$), but for the sinusoids the frequencies will be changed, and hence they are not invariant under the transformation. Thus when the units of the independent variable are not reasonably well known, the polynomial approximation is the natural one to use.

Second, the polynomial treatment was arranged to introduce many of the ideas and methods necessary for a careful examination of the problems of computing. The Fourier approach further developed many of the ideas we need because in a very real sense they are simply exponentials with a pure imaginary exponent. Thus much of the material needed for exponential approximation has already been presented.

Third, exponentials are much less used in practice than the other two, and indeed they have played a much smaller role in the history of mathematics. Thus there is less necessity to present material on this topic in a reasonably balanced presentation of numerical methods involving one independent variable.

Finally, it is part of the plan of this book to deliberately pass from giving many of the details to giving fewer as the reader matures; it should no longer be necessary to give as many details as in the earlier parts of the book.

The problems of exponential approximation using a finite number of terms can be divided as follows: first, problems in which the exponents are known in advance; second, those in which the exponents are to be determined. We examine these two in this chapter. In the following chapter we examine the use of a continuum of exponentials in the Laplace transform (which corresponds to the Fourier transform).

## 39.2 LINEAR INDEPENDENCE

Before plunging into the problem of exponential approximation, we need to prove that the functions we are going to use are linearly independent over the sample points. For polynomials this was proved by showing that the Vandermonde determinant (Sec. 14.4) was not zero if the sample points were distinct. For the Fourier series we appealed to the orthogonality, and hence the linear independence, of the functions.

Consider, first, the following sum of exponential terms.

$$f(x) = C_1 e^{\alpha_1 x} + C_2 e^{\alpha_2 x} + C_3 e^{\alpha_3 x} + \cdots + C_n e^{\alpha_n x} \qquad \alpha_i \neq \alpha_j,\ i \neq j$$

To show that they are linearly independent in any continuous interval, suppose they are dependent and that the interval for which $f(x) \equiv 0$ includes the origin. Then on successive differentiations we get

$$\begin{aligned} f'(x) &= C_1\alpha_1 e^{\alpha_1 x} + C_2\alpha_2 e^{\alpha_2 x} + \cdots + C_n\alpha_n e^{\alpha_n x} = 0 \\ f''(x) &= C_1\alpha_1^2 e^{\alpha_1 x} + C_2\alpha_2^2 e^{\alpha_2 x} + \cdots + C_n\alpha_n^2 e^{\alpha_n x} = 0 \\ &\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\ f^{(n-1)}(x) &= C_1\alpha_1^{n-1} e^{\alpha_1 x} + C_2\alpha_2^{n-1} e^{\alpha_2 x} + \cdots + C_n\alpha_n^{n-1} e^{\alpha_n x} = 0 \end{aligned}$$

At $x = 0$ the determinant of the equations for determining the coefficients $C_k$ is exactly the Vandermonde, which cannot be zero if $\alpha_i \neq \alpha_j$; hence all the $C_k$ are zero, contrary to the assumption of linear dependence.

For the case of discrete samples we use the fact that $e^{\alpha x} \neq 0$ for any finite $x$. This is the basis for an inductive proof. We therefore assume any $n - 1$ exponentials are linearly independent over any $n - 1$ distinct points. Consider now $n$ exponentials at $n$ points.

$$\sum_{k=1}^{n} C_k e^{\alpha_k x} = \left(\sum_{k=1}^{n-1} C_k e^{(\alpha_k - \alpha_n)x} + C_n\right) e^{\alpha_n x}$$

The term in parentheses is zero for $n$ values, and by Rolle's theorem the derivative

$$\sum_{k=1}^{n-1} C_k(\alpha_k - \alpha_n) e^{(\alpha_k - \alpha_n)x} = 0$$

for at least $(n - 1)$ values, contrary to the induction hypothesis. Thus $n$ exponentials are linearly independent over any $n$ distinct points.

## 39.3 KNOWN EXPONENTS

In Chap. 15 we developed a uniform method for finding formulas based on polynomial approximation. We use this same method for finding formulas which are to be accurate for a set of exponentials having known exponents.

Suppose we have some linear operator $L(f)$, such as integration, differentiation, or interpolation, and we want the answer to be exact for a set of exponentials with known coefficients (given a corresponding set of samples). For convenience we will exclude the use of derivatives. Thus we are trying to estimate $L(f)$ as a linear combination of samples of the function $f(x_i)$ such that the formula

$$L(f) = w_0 f(x_0) + w_1 f(x_1) + \cdots + w_{n-1} f(x_{n-1})$$

will be exact for the $n$ functions

$$e^{a_0 x}, e^{a_1 x}, \ldots, e^{a_{n-1} x}$$

(Often $a_0 = 0$, so that we have a constant term in the family of exponentials.)

The defining equations are, naturally,

$$m_k = \sum_{i=0}^{n-1} w_i e^{a_k x_i} \qquad k = 0, 1, \ldots, n - 1$$

where we have now written

$$m_k = L(e^{a_k x})$$

We have shown in Sec. 39.2 that the functions $e^{a_k x}$ are linearly independent over any set of $n$ distinct points $x_i$, and so the determinant of these equations is not zero, and the equations can be solved for the unknown weights $w_i$ of the formula.

As an experiment, consider the function

$$y = \frac{1}{4} + \frac{e^{-x}}{2} + \frac{e^{-2x}}{4}$$

It takes the values

| $x$ | $y$ |
|---|---|
| 0 | 1.00 |
| 1 | 0.468 |
| 2 | 0.322 |

where we are using 3-significant-digit arithmetic. Given these data, we now try to fit it by 1, $e^{-x}$, $e^{-2x}$. The equations are:

$$\begin{aligned} 1.00 &= a + 1.00\,b + 1.00c \\ 0.468 &= a + 0.368b + 0.135c \\ 0.322 &= a + 0.135b + 0.183c \end{aligned}$$

We eliminate $a$, using the top equation as the pivot,

$$\begin{aligned} 0.532 &= 0.632b + 0.865c \\ 0.678 &= 0.865b + 0.982c \end{aligned}$$

Eliminating $b$, we find that back substitution leads to

$$\begin{aligned} c &= \frac{0.032}{0.127} = 0.252 \\ b &= 0.497 \\ a &= 0.251 \end{aligned}$$

which is pretty good for 3-digit arithmetic.

### PROBLEMS

*39.3.1* Discuss the case of the given data including both function values and derivatives.

## 39.4 UNKNOWN EXPONENTS

The problem of interpolation (representation) of a function using sums of exponentials with unknown exponents is important since it arises frequently in practice and is the basis for the usual analytic-substitution approach. Prony gave a simple method for finding the exponents *when* the given data are equally spaced. The essence of Prony's method is the separation of the problem into that of first finding the exponents and then finding the coefficients. In this respect it closely resembles the method used in Gaussian quadrature (Chap. 19).

We are assuming that the desired function $f(x)$ can be written in the form

$$f(x) = A_0 e^{\alpha_0 x} + A_1 e^{\alpha_1 x} + \cdots + A_{k-1} e^{\alpha_{k-1} x}$$

for some set of given values $x = x_j$ ($j = 1, 2, \ldots, n$) which are equally spaced. It is no real restraint to suppose that $x_j = j - 1$.

Prony observed that each of the exponentials

$$e^{\alpha_i j} = (e^{\alpha_i})^j \equiv \rho_i^j \qquad i = 0, 1, \ldots, k-1$$

satisfies some fixed $k$th-order linear difference equation

$$y(j+k) + C_{k-1} y(j+k-1) + C_{k-2} y(j+k-2) + \cdots + C_0 y(j) = 0$$

with constant coefficients. The characteristic equation of this difference equation is

$$\rho^k + C_{k-1}\rho^{k-1} + C_{k-2}\rho^{k-2} + \cdots + C_0 = 0$$

and has the roots $\rho_i$. Since the individual terms satisfy the $k$th-order, linear, homogeneous difference equation, then any linear combination of the terms also satisfies the equation. In particular the original function satisfies the equation, that is,

$$f(j+k) + C_{k-1} f(j+k-1) + \cdots + C_0 f(j) = 0 \qquad j = 1, 2, \ldots, n-k$$

The assumed form of the function has $2k$ parameters, and for the unique determination of them we need $n = 2k$ samples of data. In this case we have exactly $k$ equations to determine the $C_i$ ($i = 0, 1, \ldots, k-1$). If we are to find the $C_i$, then the persymmetric determinant (Sec. 19.3)

$$\Delta = |f(j+k)|$$

must not be zero. Knowing the $C_i$, we can put them into the characteristic equation and from this determine the roots $\rho_i$, which in turn give the exponents $\alpha_i$. If the exponents are to be real, then the roots must be nonnegative. Once we have the exponents we can then use the first $k$ of the defining equations to determine the coefficients $A_j$. Just as in the Gaussian integration we are assured that these will fit the last $k$ equations also.

An examination of this process shows that it is very similar to the Gaussian integration formulas.

As an easily followed example, consider the data $k(=2)$

| $x$ | $y$ |
|---|---|
| 0 | 32 |
| 1 | 20 |
| 2 | 14 |
| 3 | 11 |

Working our way up the displayed equations, we get, in turn, the equations for the $C_i$

$$14 + 20C_1 + 32C_0 = 0$$
$$11 + 14C_1 + 20C_0 = 0$$

whose solutions are

$$C_1 = -3/2$$
$$C_0 = 1/2$$

The characteristic equation is therefore

$$\rho^2 - \frac{3}{2}\rho + \frac{1}{2} = 0$$

from which we get

$$(\rho - 1)\left(\rho - \frac{1}{2}\right) = 0$$
$$\rho = 1, \tfrac{1}{2}$$

and

$$y_n = f(n) = A_0(1)^n + A_1(\tfrac{1}{2})^n$$

The first two defining equations are

$$32 = A_0 + A_1$$
$$20 = A_0 + A_1/2$$

whose solutions are

$$A_1 = 24$$
$$A_0 = 8$$

and we have, finally,

$$y_n = f(n) = 8 + \frac{24}{2^n} = 8 + 3x2^{3-n}$$

## PROBLEMS

*39.4.1* Fit

$$y = A_1 e^{\alpha_1 x} + A_2 e^{\alpha_2 x}$$

to the data

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | 128 | 48 | 10 | 9 |

*39.4.2* Fit

$$y = A_1 e^{\alpha_1 x} + A_2 e^{\alpha_2 x}$$

to the data

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | 0 | $\frac{1}{2}$ | $\frac{3}{4}$ | $\frac{7}{8}$ |

*39.4.3* Discuss the case when $\Delta = 0$ and we cannot determine the $C_i$.

## 39.5 LEAST-SQUARES FITTING

Many times we have more data than parameters, and we wish to find a least-squares fit to the data. Again, supposing that the data points $x_j$ are equally spaced, we will come down to more than $k$ equations to determine the $C_i$. Thus we are in a position to apply the methods of least-squares solution (Chaps. 25 and 27) of the simultaneous linear equations. Having found the coefficients of the characteristic equation, we solve them for the characteristic roots and hence the exponents. Now with these known, we can again resort to a least-squares fitting of the data by the coefficients $A_j$.

The details are straightforward, though a bit messy, and need not be given here; we need to note only that there are two stages of least-squares fitting, those for the coefficients $C_i$ of the characteristic equation and those for the coefficients $A_j$ of the exponentials.

What relation this two-stage least-squares fitting bears to the direct fitting (which is often difficult to do) is open for discussion, but it is likely to be reasonably close as measured by the resulting sum of squares of the residuals.

## 39.6 PRONY'S METHOD WITH CONSTRAINTS

Just as we examined integration formulas when various, often naturally occurring, constraints were applied, we now look at Prony's method when there are constraints.

A very common constraint is that one or more of the exponents $\alpha_i$ are known (frequently $\alpha_0 = 0$).

Suppose that we have the special case (5 parameters)

$$f(x) = A_0 + A_1 e^{\alpha_1 x} + A_2 e^{\alpha_2 x}$$

When we come to the characteristic equation, which will be a cubic

$$\rho^3 + C_2 \rho^2 + C_1 \rho + C_0 = 0$$

we know that one of the roots is $\rho = 1$. This means that

$$1 + C_2 + C_1 + C_0 = 0$$

From the data we get the equations

$$\begin{aligned} f(3) + C_2 f(2) + C_1 f(1) + C_0 f(0) &= 0 \\ f(4) + C_2 f(3) + C_2 f(2) + C_0 f(1) &= 0 \end{aligned}$$

and adding the equation

$$1 + C_2 + C_1 + C_0 = 0$$

we have the three equations necessary to determine $C_0$, $C_1$, and $C_2$. Each known exponent reduces the necessary given data by 1, thus reducing the standard set of equations by 1, but by substituting each known exponent in the characteristic equation, we gain one corresponding equation. As a result, we still have the necessary number of equations.

Conditions on the coefficients tend to resemble that of Chebyshev integration if they are on unknown exponents and Ralston integration if they are on known exponents. The constraints may lead to a linear programming problem (Sec. 43.10).

## 39.7 WARNINGS

We have supposed that all will go properly in Prony's method, but surprises await the unwary. Consider, for example, the problem of fitting

$$y(x) = A_1 e^{\alpha_1 x} + A_2 e^{\alpha_2 x}$$

to the data

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | 16 | 12 | 4 | 3 |

The equations for determining the coefficients of the characteristic polynomial are

$$4 + 12C_1 + 16C_2 = 0$$
$$3 + 4C_1 + 12C_2 = 0$$

Solving these, we get

$$C_1 = 0$$
$$C_2 = -\tfrac{1}{4}$$

The characteristic equation is therefore

$$\rho^2 - \tfrac{1}{4} = 0$$

and it has the roots

$$\rho = \pm\tfrac{1}{2}$$

This produces the exponents

$$\alpha_1 = -\ln 2$$
$$\alpha_2 = \pi i + \ln \tfrac{1}{2} = \pi i - \ln 2$$

Thus, while the approximating sum of exponentials will assume the (real) given values at the sample points, between the sample points the approximating function will be a complex number! Hence, it will normally be dangerous to use for analytic substitution.

On the other hand, complex conjugates $\rho_i$ with the real part greater than zero produce complex conjugate exponents, and with the corresponding coefficients $A_i$ conjugates of each other the sum is real.

Our trouble is that we can solve the difference equation easily in the conventional form, and this leads to complex values for the interpolating function. The solution proceeds

$$y(n) = A_1(\tfrac{1}{2})^n + A_2(-\tfrac{1}{2})^n$$
$$n = 0: \qquad 16 = A_1 + A_2$$
$$n = 1: \qquad 12 = A_1/2 - A_2/2$$
$$A_1 = 20$$
$$A_2 = -4$$

Solution:

$$y(n) = 20(\tfrac{1}{2})^n - 4(-\tfrac{1}{2})^n$$

However, this form gives trouble in interpolating in the whole interval. A dodge is to write,[1] since $\cos \pi n = (-1)^n$,

$$y(n) = (\tfrac{1}{2})^n(20 - 4 \cos \pi n)$$

[1] We could also include an arbitrary term $\sin \pi n$.

Then it is easy to get a form for interpolating all $x$ values, namely

$$y(x) = (\tfrac{1}{2})^x(20 - 4 \cos \pi x)$$

While in principle we have shown how to find the exponents, things still do not always go so well in practice; sometimes it happens that the number of terms to use is not known but is to be found. An illustration of this is radioactive decay, where the terms correspond to various half-lives in the decay chain being investigated.

Let us examine the simple case of trying to distinguish between

$$Ae^{-\alpha t} \quad \text{and} \quad \frac{A}{2}e^{-(\alpha+\varepsilon)t} + \frac{A}{2}e^{-(\alpha-\varepsilon)t} = Ae^{-\alpha t}\left(\frac{e^{-\varepsilon t} + e^{\varepsilon t}}{2}\right) \qquad \alpha > 0$$

The expression in the parentheses is

$$1 + \frac{\varepsilon^2 t^2}{2} + \frac{\varepsilon^4 t^4}{24} + \cdots$$

The difference in the two forms depends on $\varepsilon^2$, and only for large $t$ can we hope to detect this amid the background noise of measurement. But for large $t$, $e^{-\alpha t}$ is small! A similar situation applies to Laplace transforms (Chap. 40).

$$f(t) = \int_0^\infty F(\sigma)e^{-\sigma t}\, d\sigma$$

Given $F(\sigma)$, it is easy to compute $f(t)$, but given $f(t)$ in the form of data, the problem of finding $F(\sigma)$ is more difficult. One of the difficulties is that the values of $F(\sigma)$ for large $\sigma$ determine the values of $f(t)$ for small $t$, and vice versa. With proper care and known limitations on $f(t)$ or with some extra information about $F(\sigma)$, the inverse transform can sometimes be found.

## 39.8 EXPONENTIALS AND POLYNOMIALS

When the overall behavior of a problem is exponential in character, the use of a polynomial times a suitable exponent often is more manageable than a sum of exponentials. Gauss-Laguerre integration

$$\int_0^\infty e^{-x} f(x)\, dx = \sum_{i=1}^{n} w_i f(x_1)$$

further illustrates this point. Since the general techniques necessary to carry out this suggestion have been developed in Part II, we do not need to discuss them further.

## 39.9 ERROR TERMS

When the formulas have been found, it is natural to ask for the error terms. The approach in Chap. 16 of Part II started with an expansion of the arbitrary function $f(x)$ in terms of the particular function $1, x, \ldots, x^{m-1}$ for which the formula was made exact. This expansion has been generalized to arbitary sets of functions by Peterson, and a treatment of it can be found in A. S. Householder's excellent (but difficult-to-read) book "Principles of Numerical Analysis" [21].

In practice the usefulness of such error terms is open to debate, and so far they have seldom been used. The error term corresponding to that developed for band-limited functions has apparently not been investigated.

# 40

## *THE LAPLACE TRANSFORM

### 40.1 WHAT IS THE LAPLACE TRANSFORM?

Given a function $F(t)$, we say that the Laplace transform $f(s)$ is

$$f(s) = \int_0^\infty e^{-ts} F(t)\, dt$$

where for the moment we think of the variable $s$ as being real, though later we will let it take on complex values. The Laplace transform is closely related to the Fourier transform, especially the so-called two-sided Laplace transform [61]

$$\int_{-\infty}^{\infty} e^{-ts} F(t)\, dt$$

For functions which vanish identically for all negative time, the two-sided transform reduces to the one-sided, or plain, Laplace transform. This class of functions is very natural to consider in many situations, especially in experiments where before starting it is reasonable to suppose that nothing happened and that the functions of interest are identically zero. Thus we regard $F(t) = 0$ for $t < 0$.

The Laplace transform has many uses, and we cannot in one short chapter develop more than a small feeling for what it is all about. Thus we must refer the reader to the many standard texts on the topic.[1]

To each function $F(t)$ there corresponds a transform $f(s)$. It is an important theorem (Lerch's theorem) that the reverse is true within functions that are zero *except* for a set of measure zero—and such exceptions are hardly likely to be relevant in practical numerical computation. Tables of Laplace transforms are often made and are widely used.

## 40.2 SOME EXAMPLES OF LAPLACE TRANSFORMS

In order to get a feel for Laplace transforms we shall find a number of simple ones.

EXAMPLE 40.2.1 $F(t) = t^n$
Then

$$f(s) = \int_0^\infty e^{-st} t^n \, dt$$

Integration by parts gives

$$f(s) = t^n \frac{e^{-st}}{-s}\bigg|_0^\infty + \frac{n}{s}\int_0^\infty t^{n-1} e^{-st} \, ds$$

$$= \frac{n}{s}\int_0^\infty e^{-st} t^{n-1} \, ds$$

and by repetition we get

$$f(s) = \frac{n!}{s^{n+1}}$$

EXAMPLE 40.2.2 $F(t) = \sin at$
Then

$$f(s) = \int_0^\infty e^{-st} \sin at \, dt$$

Standard integration tables give

$$f(s) = e^{-st}\left(\frac{-s \sin at - a \cos at}{s^2 + a^2}\right)\bigg|_0^\infty$$

$$= \frac{a}{s^2 + a^2} \qquad s > 0$$

[1] An excellent introduction is M. R. Spiegel [55]; see also W. R. LePage [36].

EXAMPLE 40.2.3 $F(t) = \cos at$
As in Example 40.2.2 we get

$$f(s) = \int_0^\infty e^{-st} \cos at \, dt$$
$$= e^{-st}\left(\frac{-s \cos at + a \sin at}{s^2 + a^2}\right)\Bigg|_0^\infty$$
$$= \frac{s}{s^2 + a^2} \qquad s > 0$$

EXAMPLE 40.2.4 $F(t) = \begin{cases} 0 & t < a \\ 1 & t > a \end{cases}$

$$f(s) = \int_0^\infty e^{-st} F(t) \, dt$$
$$= \int_a^\infty e^{-st} \, dt = \frac{e^{-st}}{-s}\bigg|_a^\infty = \frac{e^{-as}}{s}$$

**PROBLEMS**

Find the Laplace transform $F(t) \leftrightarrow f(s)$.

*40.2.1* $\sinh at \leftrightarrow \dfrac{a}{s^2 - a^2} \qquad s > |a|$

*40.2.2* $\cosh at \leftrightarrow \dfrac{s}{s^2 - a^2} \qquad s > |a|$

*40.2.3* $c_1 F_1(t) + c_2 F_2(t) \leftrightarrow c_1 f_1(s) + c_2 f_2(s)$

*40.2.4* $F(t) = \begin{cases} 1 & 0 < a \leq t \leq b \\ 0 & \text{elsewhere} \end{cases}$

*40.2.5* $F(t) = \begin{cases} 1 - t & 0 \leq t \leq 1 \\ 0 & \text{elsewhere} \end{cases}$

## 40.3 SOME GENERAL PROPERTIES OF LAPLACE TRANSFORMS

Differentiation and integration of Laplace transforms are very common, and the rules are given as follows, where we need to take the limit from the right for a discontinuity at the origin and appropriate directions at other points of discontinuity.

If

$$f(s) = \int_0^\infty e^{-st} F(t) \, dt$$

then

$$sf(s) - F(0) = \int_0^\infty e^{-st}F'(t)\,dt$$

$$s^2f(s) - sF(0) - F'(0) = \int_0^\infty e^{-st}F''(t)\,dt$$

$$\frac{f(s)}{s} = \int_0^\infty e^{-st}\left[\int_0^t F(u)\,du\right] dt$$

Using these we can find

$$\int_0^\infty e^{-st}[t^nF(t)]\,dt = (-1)^n \frac{d^n}{ds^n}f(s)$$

$$\int_0^\infty e^{-st}\left[\frac{F(t)}{t}\right] dt = \int_s^\infty f(u)\,du$$

EXAMPLE 40.3.1 Given $F(t) = \sin t$, then

$$\int_0^\infty e^{-st}\left(\frac{\sin t}{t}\right) dt = \int_0^\infty \frac{du}{u^2+1} = \arctan 1/s$$

There are two shifting theorems for the Laplace transforms.
Given

$$f(s) = \int_0^\infty e^{-st}F(t)\,dt$$

then

$$f(s-a) = \int_0^\infty e^{-st}\{e^{at}F(t)\}\,dt$$

and if

$$G(t) = \begin{cases} F(t) - a & t > a \\ 0 & t < a \end{cases}$$

then

$$e^{-as}f(s) = \int_0^\infty e^{-st}G(t)\,dt$$

Both are immediately verifiable.

We defined (Sec. 34.7) the convolution of two functions which are 0 for $x < 0$ as

$$H(t) = \int_0^t F(u)G(t-u)\,du$$

We have, similar to the Fourier transform,

$$f(s)g(s) = \int_0^\infty e^{-st}H(t)\,dt$$

All these properties are useful in practice.

## PROBLEMS

Give the details for:

*40.3.1* The first shifting theorem

*40.3.2* The second shifting theorem

*40.3.3* The convolution theorem

## 40.4 PERIODIC FUNCTIONS

We examined periodic functions in Chaps. 31 to 33 in terms of the Fourier transform. In view of their importance we again examine this class of functions, this time in terms of the Laplace transform.

If $F(t)$ has a period $T > 0$, that is, $F(t + T) = F(t)$, then we can write

$$\begin{aligned}\int_0^{\infty} e^{-st}F(t)\,dt &= \sum_{k=0}^{\infty} \int_{kT}^{k(T+1)} e^{-st}F(t)\,dt \\ &= \sum_{k=0}^{\infty} \int_0^{T} e^{-s(t-kT)}F(t-kT)\,dt \\ &= \int_0^{T} F(t) \sum_{k=0}^{\infty} e^{-st} \cdot e^{skT}\,dt \\ &= \int_0^{T} \frac{e^{-st}F(t)}{1-e^{-sT}}\,dt \qquad s > 0\end{aligned}$$

Thus we have the extra factor

$$\frac{1}{1-e^{-sT}}$$

in the integrand to compensate for the reduced range $0 \leq t \leq T$.

## 40.5 APPROXIMATION OF LAPLACE TRANSFORMS

The method of analytic substitution, which we will discuss again in Chap. 42, is quite useful in the evaluation of Laplace transforms. Suppose we have some function $F(t)$ for which there is no value in the tables of Laplace transforms that we consult, or suppose they are experimentally determined data. What can we do? We merely examine a table of transforms for family of functions $F_i(t)$ that look suitable for approximating our given function $F(t)$. If we can find a linear representation of $F(t)$ in terms of this family

$$F(t) \simeq c_1F_1(t) + c_2F_2(t) + \cdots + c_NF_N(t)$$

then using analytic substitution and the linearity property of the Laplace transform, we get our answer. (See Secs. 9.10 and 12.5 for similar examples.)

Everything seems fine. We can apply any method we please to find the approximation, exact matching at selected points, least squares, Chebyshev, or anything else that seems appropriate. Thus we have apparently thrown the problem back onto the approximation problem. And this is what we have been doing all along. We first used the class of functions $1, x, x^2, \ldots, x^{n-1}$; later we used the class of sines and cosines; and now we are looking at exponentials. But any family of functions for which we know the analytic answers can be used provided the process is linear (and the functions are linearly independent).

Our wording, however, raises the question of how it will go in practice, or in other words, how should you match the function? An examination of the Laplace transform suggests that this is not a trivial question. Consider, for example, two functions which differ far out, but are close to each other for moderate values of $t$. How will the Laplace transform "see" the difference? We weight the difference by the factor $e^{-st}$, and for moderate values of $s > 0$ this difference simply will not be seen in the accumulated sum of integrand values. What happens far out in the $t$ variable can only be seen in the transform near the origin ($s$ small). And conversely, what is near the origin will be seen far out in the transform since even when $s$ is large, it will still enter significantly in the sum that the integral computes.

We now see the fundamental trouble with the Laplace transforms. The neighborhood of infinity is transformed into the neighborhood of the origin, and vice versa. But in the time domain, where we usually make the measurements, we have a limit of how fine a spacing we can take near the origin, and we also have a limit on how long a time we can make measurements. Thus the transform is not as well determined as we could wish. Indeed, one can say with Lanczos [33] that the inversion cannot be done accurately.

However, we find in practice that both the direct and inverse processes are done. This is accomplished by making assumptions about the behavior of the function. For example, it is often convenient to assume that the transform is, or can be approximated by, a rational function with a known degree for the numerator and the denominator. With this assumption we fit (somehow) the proper rational function to the transform, and from a table we can get the original function. This is one example of what we said in the first paragraph in this section: "We merely examine a table of transforms for a family of functions $F_i(t)$ that look suitable for approximating our given function $F(t)$." We can also apply the method for $f(s)$ if we wish.

## PROBLEMS

*40.5.1* Discuss approximating $f(s) \simeq \sum_{i=1}^{N} c_i f_i(s)$.

## 40.6 COMPLEX FREQUENCIES

In practice the variable $s$ is allowed to be complex, usually with a limit on the size of the real part so that the integral

$$f(s) = \int_0^\infty e^{-st} F(t)\, dt$$

will converge. Thinking back on the underlying meaning of the integral as a limit of a sum, we are now summing terms like (for $s = x + iy$)

$$F(t) e^{-xt} e^{-iyt}$$

which can also be written as

$$F(t)\, e^{-xt} \cos yt \qquad F(t)\, e^{-xt} \sin yt$$

Thus the Laplace transform handles complex frequencies, frequencies that are sinusoidal but with an exponential decaying factor, while the Fourier transform handles only sinusoidal terms. In this sense the Laplace transform includes the Fourier transform as a special case.

The Fourier transform considers values only along a line, much as real-variable theory considers values along the real axis. The theory of complex variables, by considering values in the whole of the complex plane, both greatly limits the class of functions and at the same time greatly expands the power of the methods used. The Laplace transform with its use of the complex plane similarly gains in power. As an example, the Fourier transform of a constant 1, (or even the function sin $t$) does not technically exist for lack of convergence, but the Laplace transform, by using a slightly negative real part of $s$, has all the necessary convergence and provides a more powerful analytical tool. (One may question, however, the increased physical power.)

When we get into the complex plane, we get an inversion formula corresponding to the Fourier inversion formula. This is

$$F(t) = \frac{1}{2\pi i} \int_{\gamma - i\infty}^{\gamma + i\infty} e^{st} f(s)\, ds \qquad t > 0$$

and is sometimes called "Bromwich's integral formula."[1] The value of $\gamma$ is chosen so that the integral converges for $s$ values on this line, and hence the Cauchy integral can be closed on the left.

[1] Oliver Heaviside (1850–1925) by his formal approach initiated research on the mathematical foundations by others and is justly regarded as the "father of the field."

The complex inversion formula has great practical limitations. Generally speaking, we cannot expect to make the measurements necessary to evaluate this integral—they can only come from some theoretical assumptions and hence are not likely to lead to computing problems.

## 40.7 A FORMULA FOR NUMERICAL INTEGRATION[1]

The numerical evaluation of the Laplace transform, given equally spaced data, is often necessary. We shall find a formula, like the Hamming-Pinkham (Sec. 18.9), that uses end corrections in the form of differences.

We begin by examining integrals, over a finite range, of the form

$$I(\lambda) = \int_0^n e^{-\lambda x} f(x)\, dx$$

Given equally spaced data, which we assume is at unit spacing, we approximate the integral by the form

$$\int_0^n e^{-\lambda x} f(x)\, dx = \sum_{k=0}^{n} e^{-\lambda k} f(k) + \sum_{s=0}^{\infty} [h_s(\lambda) e^{-\lambda n} \Delta^s f(n-s) + (-1)^s m_s(\lambda) \Delta^s f(0)]$$

where the $h_s(\lambda)$ and $m_s(\lambda)$ play the role of the $q_s$ in the earlier formula. The dependence of the coefficients $h_s(\lambda)$ and $m_s(\lambda)$ on the parameter is to be expected.

As before, we begin by requiring the formula to be exact for the integrand $f(x) = 1, x, x^2, \ldots$ and get the defining equations

$$\int_0^n e^{-\lambda x} x^p\, dx = \sum_{k=0}^{n} e^{-\lambda k} k^p + \sum_{s=0}^{\infty} [h_s(\lambda) e^{-\lambda n} \Delta^s (n-s)^p + (-1)^s m_s(\lambda) \Delta^s (0)^p]$$
$$p = 0, 1, 2, \ldots$$

To get the exponential generating function, we multiply the $p$th equation by $t^p/p!$ and sum over all the equations

$$\int_0^n e^{-\lambda x} e^{tx}\, dx = \sum_{k=0}^{n} e^{-\lambda k} e^{kt} + \sum_{s=0}^{\infty} [h_s(\lambda) e^{-\lambda n} \Delta^s e^{(n-s)t} + (-1)^s m_s(\lambda) \Delta^s e^{0t}]$$

Why do we use the exponential generating function? Simply because for the operations we are going to do, integration, summation of a series, and differencing, the exponential function is left unchanged. As a result we will be able to arrange the various terms into exponential and nonexponential terms, and the terms with the exponential will carry all the dependence on $n$.

[1] See also V. I. Krylov and N. S. Skoblya [31].

We now do the indicated operations

*1* $\int_0^n e^{-(\lambda-t)x}\,dx = \dfrac{1-e^{-(\lambda-t)n}}{\lambda-t}$

*2* $\sum_{k=0}^{n} e^{-(\lambda-t)k} = \dfrac{e^{-(\lambda-t)(n+1)}-1}{e^{-(\lambda-t)}-1} = \dfrac{e^{-(\lambda-t)n}-e^{(\lambda-t)}}{1-e^{(\lambda-t)}}$

*3* $\Delta^s e^{(n-s)t} = e^{nt}(1-e^{-t})^s$

*4* $\Delta^s e^{0t} = (e^t-1)^s$

and assemble the parts

$$e^{-(\lambda-t)n}\left[-\frac{1}{\lambda-t}-\frac{1}{1-e^{(\lambda-t)}}-\sum_{s=0}^{\infty}h_s(\lambda)(1-e^{-t})^s\right]$$
$$+\left[\frac{1}{\lambda-t}+\frac{e^{(\lambda-t)}}{1-e^{(\lambda-t)}}-\sum_{s=0}^{\infty}(-1)^s m_s(\lambda)(e^t-1)^s\right]=0$$

If we now define

$$Q(t,\lambda) = -\frac{1}{\lambda-t}+\frac{1}{e^{(\lambda-t)}-1}-\sum_{s=0}^{\infty}h_s(\lambda)(1-e^{-t})^s$$
$$P(t,\lambda) = \frac{1}{\lambda-t}+\frac{1}{e^{-(\lambda-t)}-1}-\sum_{s=0}^{\infty}(-1)^s m_s(\lambda)(e^t-1)^s$$

we have

$$e^{-(\lambda-t)n}Q(t,\lambda)+P(t,\lambda)=0$$

Comparing $Q(t, \lambda)$ with $P(t, \lambda)$ we have

$$Q(t,\lambda) = P(-t,-\lambda)$$

*provided*

$$m_s(-\lambda) \equiv h_s(\lambda)$$

Thus we need only use $Q(t, \lambda) = 0$ to find the coefficients $h_s(\lambda)$ of the differences, where

$$-\sum_{s=0}^{\infty}h_s(\lambda)(1-e^{-t})^s = \frac{1}{\lambda-t}-\frac{1}{e^{(\lambda-t)}-1}$$

Again we set

$$1 - e^{-t} = v \qquad t = -\ln(1 - v)$$

to get the power series in $v$

$$\sum_{s=0}^{\infty} h_s(\lambda)v^s = \frac{-1}{\lambda + \ln(1 - v)} + \frac{1}{(1 - v)e^{\lambda} - 1}$$

The expansion of the right-hand side in powers of $v$ will yield the coefficients $h_s(\lambda)$.

As a result of a lot of tedious algebra we find Table 40.7[1] for $A_s(\lambda)$ where

$$h_s(\lambda) = (-1)^s A_s(\lambda) + \frac{e^{\lambda s}}{(e^{\lambda} - 1)^{s+1}}$$

$$m_s(\lambda) = h_s(-\lambda)$$

Thus we have the integration formula we sought.

[1] See M. M. Kaplan, "Numerical Integration with Complex Exponential Kernels," doctoral thesis, Stevens Institute of Technology, 1969, for tables and details.

Table 40.7

| $s$ | $A_s(\lambda)$ |
|---|---|
| 0 | $-\frac{1}{\lambda}$ |
| 1 | $\frac{1}{\lambda^2}$ |
| 2 | $-\left(\frac{1}{\lambda^3} + \frac{1}{2\lambda^2}\right)$ |
| 3 | $\frac{1}{\lambda^4} + \frac{1}{\lambda^3} + \frac{1}{3\lambda^2}$ |
| 4 | $-\left(\frac{1}{\lambda^5} + \frac{1}{2\lambda^4} + \frac{11}{12\lambda^3} + \frac{1}{4\lambda^2}\right)$ |
| 5 | $\frac{1}{\lambda^6} + \frac{2}{\lambda^5} + \frac{7}{4\lambda^4} + \frac{5}{6\lambda^3} + \frac{1}{5\lambda^2}$ |
| 6 | $-\left(\frac{1}{\lambda^7} + \frac{5}{2\lambda^6} + \frac{2}{\lambda^5} + \frac{15}{8\lambda^4} + \frac{137}{180\lambda^3} + \frac{1}{6\lambda^2}\right)$ |
| 7 | $\frac{1}{\lambda^8} + \frac{3}{\lambda^7} + \frac{25}{7\lambda^6} + \frac{39}{14\lambda^5} + \frac{29}{15\lambda^4} + \frac{7}{10\lambda^3} + \frac{1}{7\lambda^2}$ |

## 40.8 MIDPOINT FORMULAS

Corresponding to the midpoint integration formula we have a midpoint formula

$$\int_0^n e^{-\lambda x} f(x)\,dx = \sum_{k=0}^{n-1} e^{-\lambda(k+1/2)} f(k+1/2)$$

$$+ \sum_{s=0}^{\infty} e^{-\lambda n} h_s^*(\lambda)\Delta^s f(n-s-1/2) + (-1)^s m_s^*(\lambda)\Delta^s f(1/2)$$

and Table 40.8.

**Table 40.8**

| $s$ | $A_s^*(\lambda)$ |
|---|---|
| 0 | $-\frac{1}{\lambda}$ |
| 1 | $\frac{1}{\lambda^2} + \frac{1}{2\lambda}$ |
| 2 | $-\left(\frac{1}{\lambda^3} + \frac{1}{\lambda^2} + \frac{3}{8\lambda}\right)$ |
| 3 | $\frac{1}{\lambda^4} + \frac{3}{2\lambda^3} + \frac{23}{24\lambda^2} + \frac{5}{16\lambda}$ |
| 4 | $-\left(\frac{1}{\lambda^5} + \frac{2}{\lambda^4} + \frac{43}{14\lambda^3} + \frac{11}{12\lambda^2} + \frac{35}{128\lambda}\right)$ |
| 5 | $\frac{1}{\lambda^6} + \frac{5}{2\lambda^5} + \frac{51}{14\lambda^4} + \frac{1{,}303}{420\lambda^3} + \frac{563}{640\lambda^2} + \frac{105}{4\lambda}$ |
| 6 | $-\left(\frac{1}{\lambda^7} + \frac{77}{30\lambda^6} + \frac{111}{28\lambda^5} + \frac{4{,}057}{840\lambda^4} + \frac{69{,}079}{22{,}400\lambda^3} + \frac{10{,}459}{12{,}800\lambda^2} + \frac{567}{2{,}560\lambda}\right)$ |
| 7 | $\frac{1}{\lambda^8} + \frac{219}{70\lambda^7} + \frac{61{,}328}{11{,}760\lambda^6} + \frac{3{,}711{,}554}{419{,}440\lambda^5} + \frac{207{,}237}{156{,}800\lambda^4} + \frac{52{,}741}{12{,}260\lambda^4} + \frac{10{,}459}{44{,}800\lambda^3} + \frac{898{,}027}{313{,}600\lambda^3} + \frac{81}{2{,}560\lambda^2} + \frac{135{,}967}{179{,}200\lambda^2} + \frac{1{,}053}{5{,}120\lambda}$ |

## 40.9 EXPERIMENTAL RESULTS

Numerical experiments (by Kaplan) indicate that the formula works about as expected.

For the Laplace transform the upper limit is infinity, and for most practical problems the corresponding differences all approach zero so that the $m_s$ terms drop out.

## 40.10 FOURIER TRANSFORMS

It is immediately evident that if in the preceding formula we write $\lambda = i\omega$, and if we do sufficient algebra, then we will have formulas for both

$$\int_0^n \cos \omega t\, f(t)\, dt \qquad \text{and} \qquad \int_0^n \sin \omega t\, f(t)\, dt$$

Alternatively, we could start from scratch and use a generating-function type of derivation technique to obtain the corresponding formulas directly.

For the Fourier transform we let both limits go to infinity, and, of course, the end correction terms will drop out, leading to the trapezoid rule for integration.

The experimental comparison of the above type of formulas with the classical Filon formula (based on local quadratic approximation of the integrand, similar to Simpson's formula) shows that the above formula is superior in accuracy as well as in the more important properties of providing an estimate of both the roundoff and truncation errors. For details the reader is referred to Kaplan's thesis.[1]

[1] Op. cit.

# 41

# *SIMULATION AND THE METHOD OF ZEROS AND POLES

## 41.1 INTRODUCTION

Simulation consumes a great deal of time on many computing machines; indeed one can look at a computer as a large, flexible general-purpose device for simulating various processes. Many simulations depend mainly on the integration of ordinary differential equations, and this is the reason we have devoted a great deal of space to the subject.

On the other hand it is difficult to discuss simulation in neat classifications with nice definite formulas strung across the page. What to do in a simulation depends very much on the source of the problem, the money available, and what is going to be done with the results. Consequently, the subject apparently is not susceptible to the usual mathematical treatment.

These two conflicting aspects of the problem cause most texts to ignore the topic after giving a few methods for integrating ordinary differential equations. However, we shall not dismiss it, even though we are reduced to working around the topic with many words and few formulas.

## 41.2 SIMULATION LANGUAGES

There are many, and undoubtedly there will be many more, simulation languages applicable to various fields. Usually the descriptions of these specialized languages give a great deal of space to how to use the language, but very little to what method of integration is used, and almost no space as to *why* the particular method was chosen.

Given a short problem where the machine time is not important, most people will grab a handy simulation language and get an answer. The answers they get will often conceal any interesting aspect of the computation that might shed light on the meaning of the solution. Consider, for example, a method of integration that actually does halve and double as the accuracy requirements demand (many methods do not, and you will never know from the output in those cases that the answer is apt to be wrong). For purposes of comparison of successive runs, the output is almost always given at a regular spacing assigned by the user. This naturally conceals the detailed behavior of the solution at exactly those interesting parts where there was a great deal of structure. Yes, we wanted numbers to study, but the motto of the book also states that we want insight—insight that can often arise from the *way* the computation proceeds, rather than from the numbers. Why at this place was a small step necessary? What regions had very large steps, and what does that mean? Does this suggest some analytic approximation in this region, and does the behavior in the region of small steps suggest some very nonpolynomial-like behavior compatible with the theory? Perhaps we can find other analytic approximations that will more clearly reveal what is going on.

This is not to say that simulation languages should not be used; often their ease of use outweighs the probable gain in insight of a more careful, and initially more costly, approach.

Most simulation languages use a method of integration at least as good as Euler's modified predictor-corrector method (Sec. 22.3), though some are as crude as the simple Euler predictor method. Usually on these latter ones, the user must specify the step size to be used throughout the whole run.

Runge-Kutta methods (Sec. 24.2) are very popular because they are both accurate (hence reasonably efficient) and self-starting (hence compact coding). They have the weakness of not doing too well on the matter of monitoring their own accuracy (unless further evaluations or other features are incorporated into the method, and these are seldom done).

The better predictor-corrector methods are occasionally used, but the rumored instabilities, no self-starting properties, and general untidiness tend to make them be passed over by programmers.

When using a simulation language with its packaged routine, or indeed any packaged routine for differential equations, one should watch carefully to see how to avoid the loss of insight that these approaches tend to produce (along with their great ease of use).

When a simulation is extremely important, or when the machine time used is going to amount to enough money that it becomes a significant part of the budget, then we need to turn to the design of special methods of integration that fit the needs of the problem. As Fowler observes,[1] the methods of design and simulation should be done together, not separately. Also, we sometimes build whole simulators in hardware, with the digital computer as one component buried inside. In those cases we also want to adapt our methods of integration to fit the situation.

There are numerous published comparisons of the various methods used in a simulation.[2] Theory says that the predictor-corrector methods should do about half as many evaluations as a Runge-Kutta method of comparable accuracy, but many reports do not seem to bear this out. One can only suspect why. One possible explanation is that the predictor-corrector methods spent a lot of their time halving and doubling, because the halving and doubling constants were too close and did not allow for the simulation of noise that is so often a part of the total problem. Another obvious explanation is that the author of the comparison does not understand the newer methods and evaluates them for what they are not supposed to do rather than for what they do in fact do. Many fail to realize that step-size changes can require changing many of the coefficients (see Sec. 36.7).

Stability of the predictor-corrector method is of course another source of trouble, but if the step size has been shortened to take care of it, and if the Runge-Kutta method did not correspondingly shorten, then it is likely that the second method is in error too, but the error is not so obvious.

## 41.3 SPECIAL METHODS[3]

Usually the simulation package reduces all the equations to a single system of first-order equations. This can be very costly in machine time and include some loss of accuracy. In the very common case of second-order equations with no first derivative present there is considerable advantage in time and accuracy in going to the appropriate method (Sec. 24.3).

[1] M. E. Fowler, Numerical Methods for Use in the Study of Spacecraft Guidance and Control Systems, *IBM Scientific Computing Symposium on Digital Simulation of Continuous Systems*, pp. 47–66, 1966. See, in particular, p. 62.

[2] P. R. Benyon, A Review of Numerical Methods for Digital Simulation, *Simulation*, November, 1968, pp. 219–237, gives 75 references. H. R. Martens, A Comparative Study of Digital Integration Methods, *Simulation*, February, 1969, pp. 87–94.

[3] M. E. Fowler, A New Numerical Method for Simulation, *Simulation*, May, 1961, pp. 324–330.

The case of stiff equations,[1] which is related to the problem of widely separated time constants, is also of common occurrence. The solution to this difficulty depends on the source. Often the very short time constant that is producing the stiff part is a parasitic one that could be left out of the simulation with no serious loss in the modeling. Or, it can be replaced by an implicit equation whose derivative is taken to be zero as far as the other questions of the system are concerned. (Sec. 24.8).

Sometimes some of the equations in the simulation have solutions which are not easily approximated by polynomials (something like the stiff-equation problem), and we are forced to devise appropriate methods for integrating those particular equations in the system (Sec. 24.6).

We looked briefly at the design of integration methods from the frequency approach (Chap. 36), and we now turn to a more complete discussion (not derivation) of the approach.

## 41.4 THE FREQUENCY APPROACH AGAIN

The traditional approach of the mathematically inclined to the problem of the numerical integration of ordinary differential equations is through polynomial approximation and measuring the accuracy of the solution by a comparison of the calculated and true solutions—a comparison in the solution space. The usual approach of the electrical engineer to the same problem is through his knowledge of the frequency approach, and he measures his success by how close the Fourier, or more often the Laplace, transforms of the calculated and true solutions agree.

These two approaches are not the same, and in practice can give very different results. It is true, of course, that if one is exact, then the other is; but it is not true that if the two solutions are close to each other in the solution space, then the frequencies in each solution are close. This can be seen in Fig. 41.4.1. The first example clearly has a high-frequency component that the other function does not have, while in the second example there will be jerks at the corners of the straight-line solution that are not in the true solution, again a different frequency content.

When the solution is being used, for example, in a training simulator for supersonic flight or a space vehicle, the difference in frequency content can be serious indeed. What is wanted is a trainer that "feels" like the real thing. This is much more like the Fourier transforms being close than the solutions being

[1] M. E. Fowler and R. M. Warten, A Numerical Integration Technique for Ordinary Differential Equations with Widely Separated Eigenvalues, *IBM Journal*, September, 1967, pp. 537–543. See also Ref. 16.

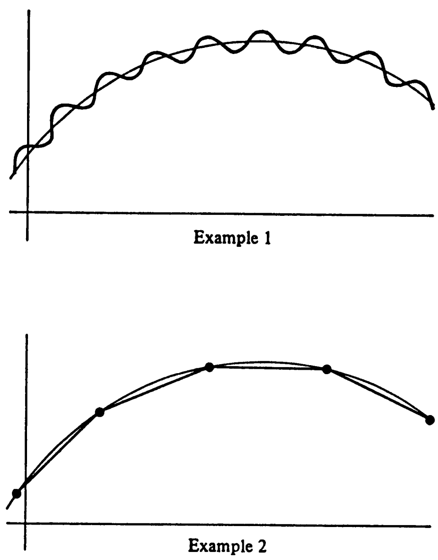


FIGURE 41.4.1

close. Indeed, it is a custom to match, not the pure sinusoid content of the solution à la Fourier, but rather the Laplace transforms over the complex plane so that the damped sinusoids will "feel proper" to the trainee.

This discussion should show why the method of zeros and poles which tries to match the Laplace transforms is in many cases much more appropriate than the conventional method of matching solutions in the solution space.

## 41.5 THE $z$ TRANSFORM

The $z$ transform is a formal analogy with other transforms and is closely related to generating functions It is very useful when dealing with sampled data systems (meaning that equally spaced samples of a function are all that are available). Let the data be (at spacing $T$)

$$f(nT) \qquad n = 0, 1, \ldots$$

where the assumption is that for negative $t$ the function is identically zero, much like the Laplace transform. The $z$ transform is defined as

$$F(z) = \sum_{n=0}^{\infty} f(nT)z^{-n} \qquad |z| > R$$

It is customary to use negative powers of $z$, though obviously a reciprocal transformation

$$z = 1/z'$$

would yield positive powers.

Like other transforms the $z$ transform is linear

$$\sum_{n=0}^{\infty}[c_1 f_1(nT) + c_2 f_2(nT)]z^{-n} = c_1 \sum_{n=0}^{\infty} f_1(nT)z^{-n} + c_2 \sum_{n=0}^{\infty} f_2(nT)$$

There is a shifting theorem

$$z[\sum f(nT)z^{-n} - f(0^+)] = \sum_{n=0}^{\infty} [f(n+1)]z^{-n}$$

The convolution is clearly the Cauchy power-series multiplication, and we have a corresponding convolution theorem

$$F_1(z)F_2(z) = \sum_{n=0}^{\infty} \sum_{n=0}^{n} f_1(kT)f_2[(n-k)T]z^{-n}$$

Thus it is easy to see that we will have all the formal properties of most transforms. We can also construct tables corresponding to functions that we can sum in closed form. The formalities need not be gone into here.[1]

The $z$ transform has been popular in some circles, and it is often used for the design of integration methods for systems of differential equations. The derivations are easier than those of the method of zeros and poles, but they lack the flexibility of the latter. Therefore, we shall not go farther into this topic, but be content to alert the reader to its existence and relation to the rest of the material.

Many of the comparisons of various methods seem to confuse the $z$ transform methods and the more general zero and poles method, so that this confusion needs to be watched carefully.

[1] See, for example, E. I. Jury [26].

PART V

# Miscellaneous

# 42

# APPROXIMATIONS TO SINGULARITIES

## 42.1 INTRODUCTION

Problems that arise in physics very often have a singularity at one end of the interval of interest, and sometimes at both ends. We have been approximating functions using (1) polynomials, (2) sines and cosines, and (3) exponentials, all of which are finite at all finite values of the argument. Thus essentially we have been solving problems with no singularities. The exceptions to this remark are that in integration we have occasionally incorporated the singularity into the kernel $K(x)$, and in differential equations we have made (by implication) transformations on the variables to eliminate the singularities; and also rational functions (Chap. 30).

What is true about problems in physics having singularities is perhaps less true in other fields of science, and even less true in engineering, but singularities do occur in practice. We therefore need to examine this important topic.

We are going to apply analytic substitution, and approximate the behavior of functions by terms which resemble the singularity. The methods we use can be adapted to any problem where we have knowledge of the nature of the solution and the appropriate family of approximating functions. When there is an